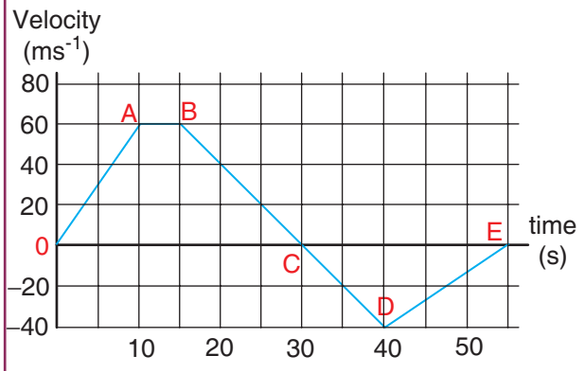
EXAMPLE 2.31

A velocity–time graph is given for a particle moving in x direction, as below



a) Describe the motion qualitatively in the interval 0 to 55 s.

b) Find the distance and displacement travelled from 0 s to 40 s.

c) Find the acceleration at t = 5 s and at t = 20 s

Solution

(a) From O to A: (0 s to 10 s)

At t = 0 s the particle has zero velocity. At t > 0, particle has positive velocity and moves in the positive x direction. From 0 s to 10 s the slope $(\frac{dv}{dt})$ is positive, implying the particle is accelerating. Thus the velocity increases during this time interval.

From A to B: (10 s to 15 s)

From 10 s to 15 s the velocity stays constant at 60 m s^{-1}. The acceleration is 0 during this period. But the particle continues to travel in the positive *x*-direction.

From B to C : (15 s to 30 s)

From the 15 s to 30 s the slope is negative, implying the velocity is decreasing. But the particle is moving in the positive x direction. At t = 30 s the velocity becomes zero, and the particle comes to rest momentarily at t = 30 s.

From C to D: (30 s to 40 s)

From 30 s to 40 s the velocity is negative. It implies that the particle starts to move in the negative x direction. The magnitude of velocity increases to a maximum 40 m s^{-1}

From D to E: (40 s to 55 s)

From 40 s to 55 s the velocity is still negative, but starts increasing from –40 m s^{-1} At t = 55 s the velocity of the particle is zero and particle comes to rest.

(b) The total area under the curve from 0 s to 40 s will give the displacement. Here the area from O to C represents motion along positive x–direction and the area under the graph from C to D represents the particle's motion along negative x–direction.

The displacement travelled by the particle from 0 s to 10 s = $\frac{1}{2}\times 10\times 60 = 300\,m$

The displacement travelled from 10 s to 15 s = 60 × 5 = 300 m

The displacement travelled from 15 s to 30 s = $\frac{1}{2}\times 15\times 60 = 450\,m$

The displacement travelled from 30 s to 40 s = $\frac{1}{2}\times 10\times(-40) = -200\,m.$

Here the negative sign implies that the particle travels 200 m in the negative x direction.

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